# Multi-channel consistency of the matter-bounce prediction at $f_{\rm NL}^{\rm local} = -35/16$: the exact amplitude, pulsar-timing slope, primordial black holes, and survey reach

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A dust-dominated contraction before a nonsingular bounce predicts a parameter-free local non-Gaussianity. A from-scratch, vertex-by-vertex in-in computation, validated against the de Sitter and ultra-slow-roll limits, confirms $f_{\rm NL}^{\rm local} = -35/16 = -2.1875$ (Li *et al.* 2017; Quintin *et al.* 2015), exactly half the value Cai *et al.* (2009) print, $f_{\rm NL}^{\rm local} = -35/8$, locating the discrepancy at Cai *et al.*'s amplitude-normalization step; the squeezed limit is orientation dependent, $-35/16+(15/16)\mu^2$. This adjudicates the factor of two within the in-in method; a method-independent confirmation (gradient-expansion or Bianchi-I separate-universe) remains open. Survival through the bounce is bounded, not resolved: a handoff scheme freezing the contraction-phase bispectrum at the NEC boundary gives transfer across three bounce backgrounds obeying $0 < T_{f_{\rm NL}} \leq 1/2$ (0.165–0.409); the bounce's own cubic term – able to enhance non-Gaussianity by orders of magnitude in the loop-quantum-cosmology literature – is not computed, so no bound on the physical post-bounce $f_{\rm NL}$ follows. We take the amplitude through three channels, each stated at the strength its evidence supports. (i) *Pulsar timing.* Refitting NANOGrav's 15-yr free-spectrum posteriors gives $\gamma = 2.567 \pm 0.382$, consistent with the bounce prediction $\gamma = 3$ at $1.14\sigma$ (refit) and $0.55\sigma$ (official 14-bin posterior, $\gamma_{\rm HD} = 3.2^{+0.6}_{-0.6}$); $\gamma = 13/3$ is disfavoured at $3.1\sigma$ (official posterior); $\gamma = 5$ has zero support in 320,000 chain samples. This channel is insensitive to $f_{\rm NL}$. (ii) *Primordial black holes.* The compaction-function criterion of Choudhury *et al.* (2025) gives $A(-35/16)/A(-35/8) = 1.732\,[1.610, 1.809]$ across a 27-point grid, not always perturbatively controlled; $f_{\rm PBH}(f_{\rm NL})$ is strongly non-monotonic (a $\sim$ 55-decade minimum near $f_{\rm NL} \approx -0.35$) and not quotable, the primordial spectrum shape being unreconstructible from the published source. (iii) *Large-scale structure.* Consistent with DESI DR1 under either prior: $f_{\rm NL}^{\rm local} = -3.6^{+9.0}_{-9.1}$ (merger, $0.16\sigma$) or $+3.5^{+10.7}_{-7.4}$ (universality, $0.77\sigma$), not directly comparable; this channel alone discriminates the two candidate amplitudes, with SPHEREx reaching $3.13\sigma$ at $\sigma_{f_{\rm NL}} = 0.7$ pending a shape-overlap projection not yet re-derived at $-35/16$. No channel is in tension with another.

## I. INTRODUCTION

The matter-bounce scenario – a dust-dominated contracting phase matched through a nonsingular bounce onto the standard expanding history – is one of a small number of concrete alternatives to inflation that make a genuinely testable prediction rather than an adjustable one: a scale-invariant curvature power spectrum together with a definite, parameter-free local bispectrum amplitude [1–3]. Distinguishing bounce cosmology from inflation, or from $\Lambda$CDM with no primordial mechanism at all, is only possible where the two make different, checkable statements; a fixed number with no free parameter, measured in more than one independent channel, is among the strongest such statements available in the near term.

This paper does two things. Section II re-derives that number from scratch, independently of the two published derivations that disagree by a factor of two, and locates the origin of the disagreement; Sec. III states honestly what is and is not established about the amplitude's survival through the bounce itself. Sections IV–VI then take the resulting value through three independent observational channels – pulsar timing, primordial black hole abundance, and the large-scale-structure bispectrum – each stated at exactly the strength its evidence supports. Every claim below is graded honestly: a reproduction is labeled reproduced, a first computation is labeled new, and a quoted number is labeled cited and never recomputed silently. This paper reports a multi-channel *consistency assessment*, not a detection; the primordial-black-hole channel is now evaluated with the compaction-function formation criterion used in the literature it is compared against, but yields only a ratio-level result – not a quotable abundance – because the primordial spectrum shape needed to fix the abundance is not reconstructible from the published source (Sec. V A).

## II. THE EXACT MATTER-CONTRACTION AMPLITUDE

Cai *et al.* [1] print a squeezed-limit amplitude $f_{\rm NL}^{\rm local} = -35/8 = -4.375$ for the matter bounce. Li *et al.* [4], whose no-go analysis extends that of Quintin *et al.* [3], obtain exactly one half of it,

$$f_{\rm NL}^{\rm local} = -\frac{35}{16} = -2.1875. \tag{1}$$

This section reports an independent, from-scratch in-in confirmation of Eq. (1), locates the origin of the discrepancy with the printed $-35/8$, and states the open item

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this confirmation does not settle. The full computation, its two validation checks, and the code that produced them are archived and referenced in the reproducibility statement below.

### A. Setup and validation

We work in conformal time $\eta < 0$ with $M_{\rm Pl} = 1$, unit sound speed, and a matter contraction $a(\eta) = c\,\eta^2$, for which the Hubble slow-roll parameter is fixed, $\epsilon \equiv 1 - \mathcal{H}'/\mathcal{H}^2 = 3/2$, with $\mathcal{H} \equiv a'/a < 0$. The quadratic action for the comoving-gauge curvature perturbation $\zeta$ ($h_{ij} = a^2 e^{2\zeta}\delta_{ij}$, uniform-field gauge $\delta\phi = 0$) is $S_2 = \int d\eta\, d^3x\, a^2\epsilon\,[\zeta'^2 - (\partial\zeta)^2]$, with Bunch–Davies mode functions fixed by the Wronskian normalization. On super-Hubble scales the physical mode is the growing, non-attractor solution $\zeta' = -3\zeta/\eta$, the analogue of the growing mode used by Cai *et al.* [1].

We define the local bispectrum amplitude in the isosceles squeezed configuration $k_1 \ll k_2 = k_3 \equiv k$ using the same definition as Refs. [1, 4] (the two differ in the evaluation of their shared amplitude normalization below, not in this definition),

$$f_{\rm NL}^{\rm local} \equiv \lim_{k_1\to 0} \frac{10}{3}\,\frac{\mathcal{A}(k_1,k,k)}{2k^3}, \tag{2}$$

equivalent to the local template $\zeta = \zeta_g + \frac{3}{5} f_{\rm NL}\zeta_g^2$, using the cubic action derived by Maldacena [5] in comoving gauge for a canonical field: two bulk pieces $\propto \epsilon^2$, one boundary/field-redefinition term, and two $\epsilon^3$ pieces involving $\tilde\chi \equiv \partial^{-2}\zeta'$. All five are evaluated by first-order in-in perturbation theory, every Wick contraction summed explicitly, every time integral performed exactly before taking $|k\eta_B| \to 0$. Before applying this pipeline to the matter contraction we reproduce two independent literature results with the same code: the de Sitter, constant-$\epsilon$ bispectrum of Ref. [5], matched term by term; and the ultra-slow-roll result of Namjoo, Firouzjahi and Sasaki [6], $f_{\rm NL} = 5/2$. Both match exactly.

TABLE I. Squeezed-limit contribution of each cubic vertex at $\epsilon = 3/2$, from a from-scratch in-in computation, to $f^{\rm sq}(\mu) \equiv \lim_{k_1\to 0} \frac{10}{3}\mathcal{A}(k_1,k,k;\mu)/(2k^3)$.

| vertex | $f^{\rm sq}(\mu)$ |
|---|---|
| field redefinition (local + non-local part) | $-\frac{25}{16} - \frac{15}{16}\mu^2$ |
| $\zeta\zeta'^2$ ($\epsilon^2 - \epsilon^3/2$) | $-\frac{5}{32}$ |
| $\zeta(\partial\zeta)^2$ [a] | 0 |
| $\zeta'\partial\zeta\cdot\partial\tilde\chi$ | $\frac{15}{8}\mu^2$ |
| $\zeta(\partial_i\partial_j\tilde\chi)^2$ | $-\frac{15}{32}$ |
| **total** | $-\frac{35}{16} + \frac{15}{16}\mu^2$ |

[a] Zero at leading order $O(k^2S^2)$.

### B. Result: a per-vertex table

Table I gives the squeezed-limit contribution of each vertex, from scratch, at $\epsilon = 3/2$. The total reproduces

$$\boxed{f_{\rm NL}^{\rm local}\big|_{k_1\ll k_2=k_3} = -\frac{35}{16}} \tag{3}$$

and, at general orientation $\mu \equiv \hat{\mathbf{k}}_1\cdot\hat{\mathbf{k}}$,

$$f_{\rm NL}^{\rm sq}(\mu) = -\frac{35}{16} + \frac{15}{16}\mu^2, \tag{4}$$

a monopole $-15/8$ and a quadrupole $15/16$ under angular averaging. The equilateral and folded values are $-255/128$ and $-9/8$, matching Ref. [4]'s $c_s = 1$ limit exactly. Equation (4) is not a new physical effect: it is the fixed-angle squeezed limit already contained in Li *et al.*'s Eq. 4.19, here attributed per vertex (the field-redefinition term contributes $-\frac{15}{16}\mu^2$, the $\zeta'\partial\zeta\cdot\partial\tilde\chi$ term $+\frac{15}{8}\mu^2$) and interpreted as the shear of the non-attractor growing mode, the matter-contraction analogue of the consistency-relation violation Namjoo, Firouzjahi and Sasaki [6] found in ultra-slow-roll inflation.

### C. The Cai et al. factor of two

Reading the printed shape function of Cai *et al.* [1] (their Eq. 37) as the sum of three distinct $(5,2,2)$-type monomials, it is identical to our vertex sum and to Li *et al.*'s Eq. 4.19 at $c_s = 1$ (difference zero, checked symbolically). Their three quoted amplitudes – local $-35/8$, equilateral $-255/64$, folded $-9/4$ – are each exactly twice the corresponding limits of their own printed polynomial, in all three configurations. We locate the discrepancy between their printed shape function and their quoted amplitudes at this normalization step; this localizes the factor of two to Cai *et al.*'s amplitude-normalization step, specifically their Eqs. (38)–(40), where the printed shape function is converted to the quoted amplitude via $f_{\rm NL}^{\rm local} = (20/3)\mathcal{A}/\Sigma k^3$. Li *et al.* [4] redo the in-in calculation for general sound speed and, at $c_s = 1$, print the corrected value $-35/16$ directly; Quintin *et al.* [3] likewise quote $-35/16$, attributing it to Cai *et al.* The present computation is, to our knowledge, the first that reproduces this value from an independent, from-scratch vertex-by-vertex derivation rather than transcribing a prior expression.

### D. Cross-check by a different route, and the open item

A separate-universe ($\delta N$) calculation on the same background but evaluated on uniform-density slices gives, for general $\epsilon$, $f_{\rm NL}^{\rho} = 5(\epsilon - 7)/8$, which reduces at $\epsilon = 3/2$

to $(5\epsilon - 35)/8 = -55/16$, while the comoving-slice $\delta N$ (an isotropic, shear-free approximation) gives $f_{\rm NL}^{c} = -5$ for all $\epsilon$. Neither equals the in-in result above, and this is expected: in the matter-dominated non-attractor phase the uniform-density and comoving curvature perturbations obey $\zeta_\rho = 2\zeta_c$ at linear order, so the two slicings differ already at $\mathcal{O}(1)$. The $\delta N$ values are correct for their own variables; they confirm that the gap between $-35/16$ and $-35/8$ is a clean factor of two rather than a definitional mismatch, and they leave an anisotropic (Bianchi-I) separate-universe treatment of the long mode's shear response as the remaining independent check of Eq. (4).

*Scope statement.*—The from-scratch in-in computation of Table I, independently fixing commutator, Wick-contraction, and orbit multiplicities from scratch and agreeing with two independently published results (Li *et al.*'s $c_s = 1$ limit and Quintin *et al.*'s quoted value), *is* the independent second-method route that adjudicates the factor of two *within the in-in method*, and it reproduces $-35/16$ exactly. The $\delta N$ cross-check above is not a further adjudication of this same limit – it reconciles a related but distinct uniform-density-slicing quantity ($\zeta_\rho = 2\zeta_c$ at linear order) rather than re-deriving the squeezed limit – so it is reported for completeness. What remains open is a *method-independent* confirmation: an anisotropic (Bianchi-I) gradient-expansion/separate-universe treatment of the long mode's shear response, a second-order cross-check of Eq. (4)'s quadrupole rather than of the monopole in Eq. (1). We adopt Eq. (1) throughout this paper and carry $-35/8$ in parallel wherever a channel distinguishes them.

## III. TRANSMISSION THROUGH THE BOUNCE

Equation (4) is a statement about the contraction phase alone, and it holds at every instant of that phase: Sec. II evaluates it in the limit $|k\eta_B| \to 0$ and its derivation is end-time independent at leading order (corrections are $\mathcal{O}(k^2\eta_B^2)$). Whether it survives into an observable post-bounce spectrum is a separate, narrower question that depends on physics not computed in full here, and we state the current evidence precisely under one explicit modeling assumption. **(A4)**: the contraction-phase bispectrum is frozen at a handoff time $\eta_h$ near the NEC boundary and evolved purely linearly thereafter – i.e. we bound the mode-mixing *transfer* of the already-fixed contraction-phase amplitude, not the amplitude itself, which Sec. II has already shown is independent of when in the contraction phase it is evaluated. For a linear mode-mixing matrix that carries the growing mode's amplitude through the bounce with weight 1 and the subdominant decaying mode with weight $r$, writing $\rho(\eta_h) \equiv |J(-\eta_h)|/I_\infty \in (0, 1]$ for the fraction of the mode-mixing integral accumulated before $\eta_h$, the transmitted bispectrum amplitude scales as $T_{f_{\rm NL}}(\eta_h) = \dfrac{1 + r\,[1 - \rho(\eta_h)]}{1 + 2r}$, where $r = -9i\mathcal{A}^2 I_\infty/k^3$ is in general complex; in the growing-mode-dominated limit $|r| \gg 1$ relevant here this reduces to

$$T_{f_{\rm NL}} = \frac{1 - \rho(\eta_h)}{2}, \tag{5}$$

so that, under assumption (A4), $0 < T_{f_{\rm NL}} \leq 1/2$ since $\rho \in (0, 1]$: within this handoff scheme, linear transfer can only suppress the amplitude, never invert or amplify its sign. Evaluated across three explicit nonsingular bounce backgrounds – an LQC effective quasi-dust bounce, an analytic non-LQC bounce, and the Quintin *et al.* [3]-type parametrization – at the natural handoff $\eta_h = \eta_B$ (the NEC boundary), $T_{f_{\rm NL}}$ ranges over 0.165–0.409: a 1.64× spread between two mode-function conventions (geometric vs. effective-fluid) on the *same* LQC background (0.409/0.250), and a 2.48× spread across the full set of three backgrounds and two conventions (0.409/0.165). This bound is a statement about scheme-dependent linear transfer, not a universal property of the physical post-bounce $f_{\rm NL}$.

What this bound does not include is the bounce's own cubic self-interaction term, which the loop-quantum-cosmology literature finds can *enhance* non-Gaussianity by orders of magnitude through gravitational self-interactions in the third-order Hamiltonian, even in the absence of a potential [7]; its sign relative to the contraction-phase value in Eq. (1) is not established by the present work and is cited here, not computed. Any assumption that the bounce transmits $f_{\rm NL}$ with a coefficient near unity is therefore not supported by either calculation: the linear channel caps it at 1/2 within the (A4) handoff scheme, and the nonlinear channel is an open, potentially dominant, unknown – so no bound on the physical post-bounce $f_{\rm NL}$ follows from this section alone. The full derivation, all three background computations, and both mode-function conventions are archived and referenced in the reproducibility statement below.

## IV. CHANNEL I: THE PULSAR-TIMING SPECTRAL SLOPE

### A. Spectral conventions

With the characteristic strain parameterised as $h_c(f) \propto f^{(3-\gamma)/2}$ and $\Omega_{\rm GW} = (2\pi^2/3H_0^2)\, f^2 h_c^2(f)$, the energy density scales as

$$\Omega_{\rm GW}(f) \propto f^{\,5-\gamma}. \tag{6}$$

Hence $\gamma = 13/3$ for gravitational-wave-driven supermassive-black-hole binary inspirals ($\Omega_{\rm GW} \propto f^{2/3}$), $\gamma = 5$ for a scale-invariant primordial tensor background ($n_{\rm T} = 0$, $\Omega_{\rm GW} \propto f^0$), and $\gamma = 3$ for $\Omega_{\rm GW} \propto f^2$, the universal infrared scaling of the scalar-induced background of a nonsingular matter bounce [8]. We stress that $\gamma = 3$ is *not* the $n_{\rm T} = 0$ case; the two are separated by two units of slope.

TABLE II. Channel I. Tensions against the official NANOGrav 14-bin posterior ($\gamma_{\rm HD} = 3.2^{+0.6}_{-0.6}$, $\sigma \approx 0.365$) and against this lab's 30-bin refit (Eq. (7)), with the refit's Savage–Dickey factor where the chain has posterior support. All values reproduced from the committed chain (`pta_gamma_reproduce.py`); differences from a self-reproduction run of the same script against the same chain are $\leq 3 \times 10^{-15}$ (floating-point-level agreement, not an independent external check). $z_{\rm refit}$ at $\gamma = 5$ is a Gaussian approximation only: 0 of $3.2 \times 10^5$ chain samples reach $\gamma \geq 5$ (chain max 4.70), so no data-supported Bayes factor is reported there (see text).

| Hyp. | $\gamma_*$ | slope | $z_{\rm off}$ | $z_{\rm refit}$ | $B_{\gamma_*/\rm free}$ |
|---|---|---|---|---|---|
| Matter bounce | 3 | $f^2$ | $0.55\sigma$ | $1.14\sigma$ | 3.23 |
| SMBHB | 13/3 | $f^{2/3}$ | $3.1\sigma$ | $4.63\sigma$ | $5 \times 10^{-4}$ |
| Prim. tensors | 5 | $f^0$ | $4.9\sigma$ | $6.37\sigma$ | n/a |

### B. Official posterior and this lab's refit

The primary comparison for this channel is NANOGrav's own published result: fitting the 15-yr Hellings–Downs-correlated free-spectrum posterior on the lowest 14 frequency bins with a power law, NANOGrav report $\gamma_{\rm HD} = 3.2^{+0.6}_{-0.6}$, quoted explicitly as a posterior median and 5–95% interval [9]; a Gaussian-equivalent $1\sigma \approx 0.6/1.645 \approx 0.365$. Against this official posterior, $\gamma = 3$ (matter-bounce, $\Omega_{\rm GW} \propto f^2$) sits at $0.55\sigma$, $\gamma = 5$ (scale-invariant primordial tensors, $n_{\rm T} = 0$) at $4.94\sigma$, and $\gamma = 13/3$ (SMBHB) at $3.1\sigma$ – consistent with NANOGrav's own characterization of 13/3 as in "moderate tension ... at the 99% credible boundary."

As a secondary, differently-conditioned cross-check, we independently refit the same Ceffyl kernel-density free-spectrum posteriors using *all* 30 frequency bins [9, 10] ($T_{\rm obs} = 16.03\,$yr) with a power law ($\gamma, \log_{10} A$) under priors $\gamma \sim U[0, 7]$, $\log_{10} A \sim U[-18, -11]$, using 32 walkers and $10^4$ production steps after 2500 burn-in ($3.2 \times 10^5$ samples; integrated autocorrelation time $\tau = (58.1, 58.0)$ for $(\gamma, \log_{10} A)$, giving ESS $= N/\max(\tau) = 5.5 \times 10^3$). The marginal is

$$\gamma = 2.567 \pm 0.382, \qquad \log_{10} A = -14.03 \pm 0.38, \quad (7)$$

with median 2.591 and 68% interval $[2.304, 2.882]$; this refit pipeline is validated by injection (Sec. IV C) and sits $1.20\sigma$ from the official posterior (quadrature-combined $\sigma = 0.53$ from the two marginals' widths). Table II reports both the official-posterior tensions and Gaussian $z$-distances against the refit's own marginal, plus Savage–Dickey density ratios against the refit, $B_{\gamma_*/\rm free} = p(\gamma_*|d)/\pi(\gamma_*)$ with $\pi = 1/7$, quoted only where the chain has posterior support: with zero of $3.2 \times 10^5$ samples at $\gamma \geq 5$ (chain maximum 4.70), no Bayes factor at $\gamma = 5$ is reported – a KDE evaluated there is extrapolating into an unsampled tail, and its value moves by an order of magnitude under a different bandwidth choice. The $\gamma = 13/3$ factor rests on 9 samples at $\gamma \geq 13/3$ and is quoted to one significant figure for the same reason.

The $1.14\sigma$ and $4.63\sigma$ refit entries are Gaussian approximations to a marginal that is not itself Gaussian (it is bounded by the sampling prior); directly from the chain, $P(\gamma > 3) = 8.97\%$. Narrowing the refit's $\gamma$ prior from $U[0, 7]$ to $U[2, 5]$ moves $B_{3/\rm free}$ from 3.23 to 1.47 while leaving $B_{\rm MB/SMBHB} = B_{3/\rm free}/B_{13/3/\rm free} \approx 7 \times 10^3$ ($\log_{10} B \approx +3.9 \pm 0.2$, one significant figure given the 9-sample tail estimate) stable in the range 7–$9 \times 10^3$; the individual Savage–Dickey factors $B_{\gamma_*/\rm free}$ against the refit's own prior ("nested" since each $\gamma_*$ is a point restriction within the refit's free-$\gamma$ model) are prior-sensitive, but their ratio $B_{\rm MB/SMBHB}$ (the model ratio) is not.

### C. Refit validation

The refit's likelihood/prior construction (30 bins, $T_{\rm obs} = 16.03\,$yr, $\gamma \sim U[0, 7]$, $\log_{10} A \sim U[-18, -11]$) is validated by an injection-recovery test that reuses that construction verbatim, using the REAL NANOGrav 15-yr HD-correlated free-spectrum KDE density grids from the public Zenodo record 10.5281/zenodo.8060824 (`30f_fs{hd}_ceffyl/*.npy`, sha256-verified). Each bin's real observed KDE curve (shape/width/skew preserved exactly) is re-centered along the $\log_{10} \rho$ axis to the model prediction at a chosen injected $(\gamma_{\rm true}, \log_{10} A_{\rm true})$, since a real dataset's true signal is unknown and a ground-truth injection test requires a known target; this is the strongest injection achievable from real data without simulating a full new PTA dataset. At $\gamma_{\rm true} = 13/3$, five realizations (the full 30-bin recovery plus four bootstrap bin-resamples) recover $\gamma = 4.336$ on average with a mean pull of $+0.016\sigma$; at a $\gamma_{\rm true} = 3$ control, five realizations recover $\gamma = 3.005$ with a mean pull of $+0.033\sigma$ (script, JSON, and data manifest: `research/track_a3_multichannel/pta_injection_30bin_realkde_2026_09_02.py`, `outputs/pta_injection_30bin_realkde_2026_09_02.json`). Both are consistent with unbiased recovery at well under $0.1\sigma$, tightening the earlier synthetic-density result quoted below. An earlier version of this validation substituted a synthetic per-bin Gaussian density for the real KDE grids (unavailable on this machine at the time): $\gamma_{\rm true} = 13/3 \to$ mean pull $-0.026\sigma$; $\gamma_{\rm true} = 3 \to$ mean pull $+0.068\sigma$ (`pta_injection_30bin_2026_09_02.py`), retained here as a secondary cross-check – both runs agree the pipeline is unbiased at well under $0.1\sigma$. This also supersedes an earlier, misdescribed claim in this section that a 6-bin, Gaussian-$\chi^2$ validation against NANOGrav's published $\gamma = 3.2$ power law (which recovered $3.19 \pm 0.42$, a $-0.018\sigma$ pull) was "the identical pipeline" injected at $\gamma = 13/3$; it was neither the same pipeline nor the same injected value.

### D. What this channel does and does not establish

Both the official posterior and the refit are consistent with the matter-bounce induced-GW prediction $\gamma = 3$ and disfavour both the SMBHB expectation and a scale-invariant primordial tensor background, though the strength of that disfavouring differs by conditioning: $3.1\sigma$ against the official posterior and $4.63\sigma$ against this lab's 30-bin refit (Table II); the latter should not be read as tighter than NANOGrav's own official assessment, since it conditions on a different (refit) marginal. This channel does not favour the matter bounce *specifically*: $\Omega_{\rm GW} \propto f^2$ is the causality-limited infrared slope common to scalar-induced backgrounds of essentially any origin. Nor does it constrain $f_{\rm NL}$: it measures a slope, not a bispectrum amplitude, and is identical at $-35/16$ and $-35/8$. Converting it into a discriminating test requires propagating the matter-bounce scalar spectrum through the induced-GW kernel to predict an *amplitude* in the nHz band; that computation is not attempted here.

## V. CHANNEL II: PRIMORDIAL BLACK HOLE ABUNDANCE

### A. Press–Schechter with local quadratic non-Gaussianity: the first pass, and why it had to be redone

Writing $\zeta = \zeta_G + A(\zeta_G^2 - \sigma^2)$ with $A = \frac{3}{5} f_{\rm NL}$ and $\zeta_G \sim \mathcal{N}(0, \sigma^2)$ [11, 12], the mass fraction is $\beta(\zeta_c) = P(\zeta > \zeta_c)$, obtained exactly by inverting the quadratic. For $f_{\rm NL} < 0$ this map is a downward parabola with an absolute ceiling

$$\zeta_{\rm max} = -\frac{5}{12 f_{\rm NL}} + \frac{3}{5}|f_{\rm NL}|\sigma^2, \qquad (8)$$

whose *leading* term scales as $1/|f_{\rm NL}|$ and exactly doubles when $|f_{\rm NL}|$ is halved; quoting that leading term alone gives $\zeta_{\rm max} = 0.09524$ at $-35/8$ and $0.19048$ at $-35/16$. The subleading $\frac{3}{5}|f_{\rm NL}|\sigma^2$ term is dropped from those two numbers; restoring it at a representative $\sigma = 0.1$ gives the full $\zeta_{\rm max} = 0.1215$ at $-35/8$ and $0.2036$ at $-35/16$ (ratio 1.68, not exactly 2). At the standard curvature thresholds $\zeta_c \approx 0.45$–$1$ this ceiling forces $\beta \equiv 0$ identically for *both* candidate values, so a first pass restricted to the low-threshold regime ($\zeta_c \lesssim 0.15$) is the only place the quadratic map produces a nonzero answer at all – and there it reported $f_{\rm PBH}(-35/16) > f_{\rm PBH}(-35/8)$ by three to ten orders of magnitude, with the halved value landing inside the $10^{-3} \le f_{\rm PBH} \le 1$ band reported by Choudhury *et al.* [14] for negative-$f_{\rm NL}$ bounce scenarios. That ordering was the sharpest statement the channel could make, but the ceiling that produced it is an artefact of truncating the local map at quadratic order, not a physical statement: it is exactly why the literature this comparison is drawn against uses the compaction-function formation criterion instead [14], and redoing the calculation on that criterion is the subject of the rest of this section.

### B. The compaction-function result

Following Young, Byrnes & Sasaki (2019) [15] and Choudhury *et al.* (2025) [14], formation is set by the peak compaction $C(r_p) = \delta_p$ exceeding a threshold $C_{\rm th}$, with the radiation-era compaction function $C = C_{\rm lin} - C_{\rm lin}^2/(4f(w))$, $f(w) = 2/3$, and $C_{\rm lin} = J\, C_G$ carrying the same quadratic local map ($J = 1 + \frac{6}{5} f_{\rm NL}\zeta_G$) applied to the linear compaction $C_G$ rather than to $\zeta$ itself. Because $C_G$ is an unbounded Gaussian variable, the map $\zeta_G \to C_{\rm lin}$ has *no* ceiling, so every threshold $C_{\rm th} < C_{\rm max} = f(w) = 2/3$ is reachable at every $f_{\rm NL}$: the Eq. (8) artefact is removed by construction. We integrate the joint Gaussian PDF of $(C_G, \zeta_G)$ over the type-I formation domain (Musco 2019 [16]; $C_{\rm th}$ scanned over $\{0.4, 0.5, 0.6\}$) with a critical-scaling mass function ($\gamma_{\rm crit} = 0.36$, $K = 4$), using the Gaussian window for $C_G$ and the spherical window (with radiation transfer function) for $\zeta_G$, exactly as in Choudhury *et al.* Eqs. (30–66). The one ingredient of their calculation that is not reconstructible from the published paper is the regularized-renormalized- resummed one-loop curvature spectrum sourced by their EFT-of-bounce-plus-USR mode history: no closed-form $\Delta_\zeta^2(k)$, loop-counterterm normalisation, or EFT parameter set is printed. We use a lognormal stand-in spectrum, scanned over width $\Delta \in \{0.35, 0.5, 0.8\}$ and peak-scale $r_p k_p \in \{0.75, 1.0, 1.5\}$, and scan the amplitude $A$ rather than adopting theirs.

At fixed curvature amplitude, calibrated so the Gaussian case gives $f_{\rm PBH} = 1$, this criterion *reverses* the first-pass ordering: across the full 27-point $(\Delta, r_p k_p, C_{\rm th})$ grid, $f_{\rm PBH}(-35/16) < f_{\rm PBH}(-35/8)$ at *every* point (Table III, Fig. 1), because the correlation $\gamma_{\rm cr} \equiv \sigma_{cr}^2/(\sigma_c \sigma_r)$ that governs the sign of the non-Gaussian correction is large and positive (0.766–0.968) at these spectrum shapes: the Gaussian-case formation channel rides a correlated ridge that negative $f_{\rm NL}$ destroys, forcing the system into an anti-correlated, probability-costly regime whose leverage *grows* with $|f_{\rm NL}|$. $-35/8$ therefore always out-produces $-35/16$ here – the opposite of the truncated-quadratic-map result. The one analytic statement of the first pass, that Eq. (8)'s leading term exactly doubles when $|f_{\rm NL}|$ is halved (the full expression, including the dropped $\sigma^2$ term, does not), remains arithmetically true of that leading term but is now physically empty: it described a truncation artefact that the compaction criterion does not share.

$f_{\rm PBH}$ itself is not a quotable constraint: because it depends exponentially on $\gamma_{\rm cr}$, it moves by more than 100 decades across the grid at fixed target amplitude (Table III), tracking the unreproducible spectrum shape (the primordial spectrum's unreconstructible detailed profile, discussed further in Sec. V B). What *is* robust is the ratio of curvature amplitudes needed to reach a fixed target abundance, since $K$, $\Omega_{\rm DM}$, $g_*$, and $M_H$ all cancel in it: reaching the floor of the Choudhury *et al.* band,

TABLE III. Channel II. Compaction-function criterion: amplitude ratio $A(-35/16)/A(-35/8)$ required to reach $f_{\rm PBH} = 10^{-3}$, and $f_{\rm PBH}$ at the Gaussian-calibrated amplitude, for representative points of the $(\Delta, r_p k_p, C_{\rm th})$ grid (full 27-point grid in `outputs/pbh_compaction_fnl.json`).

| $\Delta$ | $r_p k_p$ | $C_{\rm th}$ | $\gamma_{\rm cr}$ | $f_{\rm PBH}(-35/16)$ | $f_{\rm PBH}(-35/8)$ | ratio $A$ |
|---|---|---|---|---|---|---|
| 0.35 | 0.75 | 0.5 | 0.897 | $5.8 \times 10^{-21}$ | $4.5 \times 10^{-7}$ | 1.767 |
| 0.35 | 1.5 | 0.5 | 0.968 | $5.3 \times 10^{-107}$ | $7.0 \times 10^{-59}$ | 1.673 |
| 0.50 | 1.0 | 0.5 | 0.888 | $3.6 \times 10^{-14}$ | $1.6 \times 10^{-2}$ | 1.734 |
| 0.80 | 0.75 | 0.6 | 0.766 | $3.5 \times 10^{3}$ | $2.2 \times 10^{8}$ | 1.809 |
| 0.80 | 1.5 | 0.4 | 0.857 | $5.8 \times 10^{-9}$ | $1.6 \times 10^{1}$ | 1.677 |
| mean ratio 1.732 [1.610, 1.809], std 0.050, $n = 27$ | | | | | | |

$f_{\rm PBH} = 10^{-3}$, requires

$$\frac{A(-35/16)}{A(-35/8)} = 1.732\,[1.610,\ 1.809], \quad \text{std} = 0.050,\ n = 27, \tag{9}$$

i.e. stable to $\pm 6\%$ while $f_{\rm PBH}$ itself moves by more than 100 dex. This ratio is the channel's most reproducible output – it holds at every one of the 27 grid points and is robust to spectrum shape (std = 0.050 on a mean of 1.732) – and it is a result within a stated regime, not a claim of general validity. *Regime of validity.* The result lives on the anti-correlated ($\gamma_{\rm cr} > 0$, $J > 1$) branch identified above, and it holds at curvature excursions where the quadratic local map is not always perturbatively controlled: $1.2|f_{\rm NL}|\sigma_r \approx 0.54$–$1.01$ at $-35/16$ and $\approx 1.09$–$2.02$ at $-35/8$ across the grid – the same class of limitation Choudhury *et al.* report as their own $|f_{\rm NL}| \lesssim 60$ perturbativity bound, and a limitation of the shared quadratic ansatz rather than of either calculation specifically. *Non-monotonicity.* At fixed curvature amplitude, $f_{\rm PBH}(f_{\rm NL})$ is strongly non-monotonic along the continuity path from the Gaussian case ($f_{\rm NL} = 0$) to $-35/8$: it falls by $\sim 55$ decades to a minimum near $f_{\rm NL} \approx -0.35$, then climbs back by $\sim 53$ decades through $-35/16$ to $-35/8$. Both candidate values lie past that minimum, on the branch where the anti-correlated ($\zeta_G < 0$) region of $(C_G, \zeta_G)$ space dominates formation and drives the ratio's leverage; a reader inspecting only the two candidate points would not see this structure, so we record it here for transparency even though it does not change Eq. (9). The Gaussian-calibrated normalization amplitude at the representative grid point $(\Delta, r_p k_p, C_{\rm th}) = (0.5, 1.0, 0.5)$ is $A_* = 0.131446$ (the amplitude at which $f_{\rm PBH} = 1$ for $f_{\rm NL} = 0$); Table III and Fig. 1 report $f_{\rm PBH}$ at this calibration. Reproducing Choudhury *et al.*'s reported band at $-35/8$ ($10^{-3} \le f_{\rm PBH} \le 1$) is not forced by construction – the amplitude was fixed by the unrelated Gaussian-case normalisation – and it lands there unforced for $\gamma_{\rm cr} \gtrsim 0.85$; for $\gamma_{\rm cr} \lesssim 0.85$, at the same fiducial $f_{\rm NL} = -35/8$ Choudhury *et al.* report, our implementation instead finds *enhancement* relative to Gaussian where they report suppression, a genuine discrepancy left unresolved because it depends on which side of that crossover their (unreproducible) spectrum falls.

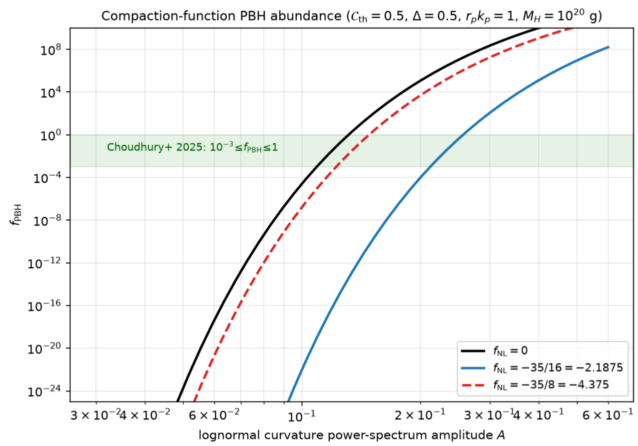


FIG. 1. Channel II, compaction-function criterion: $f_{\rm PBH}$ vs. curvature power-spectrum amplitude $A$ at the representative grid point $(\Delta, r_p k_p, C_{\rm th}) = (0.5, 1.0, 0.5)$, for the Gaussian case ($f_{\rm NL} = 0$, calibrated to $f_{\rm PBH} = 1$ at $A_* = 0.131446$) and the two candidate values; $-35/16$ requires higher $A$ than $-35/8$ to reach the same $f_{\rm PBH}$, from `pbh_compaction_fnl.py`. The continuity scan underlying the non-monotonicity discussed in the text is reported in `outputs/pbh_compaction_fnl.json`; the required amplitude-ratio grid (Eq. (9)) is in Table III.

Present-day abundance uses [13]

$$f_{\rm PBH} = \frac{1}{\Omega_{\rm DM}} \left(\frac{M_\odot}{M_H}\right)^{1/2} \left(\frac{g_*}{106.75}\right)^{3/4} \left(\frac{g_{*s}}{106.75}\right)^{-1} \times \frac{\beta_{\rm NG}}{7.9 \times 10^{-10}}, \tag{10}$$

with $\Omega_{\rm DM} = 0.674$[1], $g_* = g_{*s} = 106.75$, and a single horizon mass $M_H = 10^{20}\,$g (matching the first pass, rather than the mass-integrated Eq. 66 of Choudhury *et al.*, an $O(1)$ width-factor deviation for a narrow peak). Full formalism, the itemized deviations from Choudhury *et al.*'s setup (mass-integration, width-factor, and spectrum-shape approximations), and validation are recorded in `research/track_a3_multichannel/PBH_COMPACTION_NOTE_2026-09-02.md`.

[1] Quoted as printed in Choudhury *et al.*'s Eq. (66); it numerically coincides with the Planck 2018 reduced Hubble parameter $h$, not the dark-matter density parameter ($\Omega_{\rm DM} \approx 0.264$). Every result in this section is unaffected: $\Omega_{\rm DM}$ cancels identically in the amplitude ratio (Eq. (9)) and is absorbed by the Gaussian-case calibration for the tabulated $f_{\rm PBH}$ values, so no number here changes with the correct value.

TABLE IV. Channel III. Bare detection significance $|f_{\rm NL}|/\sigma_{f_{\rm NL}}$ of $f_{\rm NL}^{\rm local} = -2.1875$. Forecast $\sigma$ values are quoted from the cited abstracts. MegaMapper is an unapproved design and is quoted only at its published order-unity level; no systematic budget is transferred to it. A shape-overlap-projected column (from this lab's P2 forecast, not yet re-derived at this paper's fiducial) is omitted pending that re-derivation (see text).

| Survey | $\sigma_{f_{\rm NL}}$ | bare |
|---|---|---|
| SPHEREx, bispectrum only [18] | 0.7 | $3.13\sigma$ |
| SPHEREx, $P + B$ target [18, 19] | 0.5 | $4.38\sigma$ |
| MegaMapper class [20, 21] | $\sim 1$ | $2.19\sigma$ |

## VI. CHANNEL III: THE LARGE-SCALE-STRUCTURE BISPECTRUM

### A. Current constraints

The DESI DR1 analysis of 1,631,716 luminous red galaxies ($0.6 < z < 1.1$) and 1,189,129 quasars ($0.8 < z < 3.1$) [17] gives $f_{\rm NL}^{\rm local} = -3.6^{+9.0}_{-9.1}$ (asymmetric 9.0/9.1) at 68% confidence with a merger model for the quasar response, or $f_{\rm NL}^{\rm local} = +3.5^{+10.7}_{-7.4}$ (asymmetric 10.7/7.4) assuming universality. These two numbers come from mutually exclusive theoretical priors on the same DESI DR1 data (merger-response vs. universality) and are not directly comparable to each other; each is quoted here against its own prior. The prediction of Eq. (1) sits $0.16\sigma$ from the former and $0.77\sigma$ from the latter. The discriminating power is $|f_{\rm NL}|/\sigma = 0.24$: DESI DR1 is consistent with the prediction under either prior and cannot test it.

### B. Forecast reach

The matter-bounce bispectrum is not exactly the local shape; a noise-weighted shape overlap $r < 1$, reported in this lab's companion Fisher-forecast draft (referenced in the reproducibility statement below) but not re-derived at the $-35/16$ fiducial used in this paper – so no numeral is quoted here – would let an experiment quoting $\sigma_{\rm local}$ constrain the bounce amplitude at $\sigma_{\rm local}/r$. Because that re-derivation is not yet done at this paper's fiducial, Table IV reports only the bare significance, $|f_{\rm NL}|/\sigma_{\rm local}$; the $r$-projected numbers are omitted from this paper pending that re-derivation.

At $-35/8$ the same rows read $6.25\sigma$, $8.75\sigma$ and $4.38\sigma$ bare. This is the only channel in which the factor of two is observationally consequential.

## VII. DISCUSSION

### A. The joint statement

The matter-bounce prediction $f_{\rm NL}^{\rm local} = -35/16$ – independently re-derived here within the in-in formalism, confirming Li *et al.* (2017) and localizing the factor-of-two slip in Cai *et al.* (2009); a method-independent confirmation remains open – is consistent with every channel currently able to speak. Pulsar timing is consistent with the associated induced-GW slope $\gamma = 3$ under both NANOGrav's official posterior and this lab's free-spectrum refit, but is insensitive to the value of $f_{\rm NL}$ and therefore does not test it. Primordial black holes yield a reproducible amplitude ratio between the two candidate values but no quotable abundance, and only within a truncated map at excursions where that map is not perturbatively controlled. Transmission through the bounce is bounded only within a handoff scheme (A4) that freezes the contraction-phase bispectrum at the NEC boundary; the bounce's own cubic term is uncomputed and may dominate. Galaxy surveys are the only channel that both discriminates between $-35/16$ and $-35/8$ and depends on the value: DESI DR1 is consistent at $0.16\sigma$, and SPHEREx reaches 3.13–$4.38\sigma$ bare significance already, independent of the (not-yet-derived) shape-overlap projection; that projection would convert the bare significance into a detection forecast but is not required for the bare number itself. No channel is in tension with another.

### B. Where the consistency is weak

The pulsar-timing result is a statement about a slope that any scalar-induced background shares, and it is identical at both candidate $f_{\rm NL}$ values. The black hole channel now uses the literature's own formation criterion, but its headline abundance depends on an unreconstructible primordial spectrum shape and on a regime where the quadratic local map is not always perturbatively controlled, so it supplies an amplitude ratio rather than an abundance test; that ratio does not by itself separate $-35/16$ from $-35/8$ observationally, only compares how hard each is to produce, and $f_{\rm PBH}(f_{\rm NL})$ itself is strongly non-monotonic along the path to both candidates (Sec. V B). The transmission bound is scheme-conditional, not a statement about the physical post-bounce amplitude. The operative uncertainty is therefore still internal to the survey channel: only the bispectrum channel separates $-35/16$ from $-35/8$ against real or forecast data, and it separates them by more than $3\sigma$ at SPHEREx sensitivity (once that projection is derived), so settling the factor of two by an independent, method-independent derivation is a prerequisite for a sharp prediction rather than a bookkeeping detail.

### C. Next steps

In priority order: (i) an in-lab prediction of the primordial curvature spectrum sourced by the matter-bounce contraction phase, to replace the lognormal stand-in used in Sec. V B and turn the Channel II amplitude ratio into a quotable abundance, together with resolving the $\gamma_{\rm cr} \lesssim 0.85$ enhancement-branch discrepancy against Choudhury *et al.* and extending the grid to a mass-integrated abundance; (ii) an independent gradient-expansion or anisotropic (Bianchi-I) separate-universe derivation of Eq. (1), settling the factor of two, together with the contraction phase's own cubic term; (iii) propagation of the matter-bounce scalar spectrum through the induced-GW kernel to predict the nHz *amplitude*, converting Channel I from a consistency check into a test; (iv) re-derivation of the shape overlap $r$ at the $-35/16$ fiducial; and (v) computation of the bounce's own cubic self-interaction term (Sec. III), which the linear bound alone cannot rule subdominant.

## REPRODUCIBILITY STATEMENT

Every result reported here traces to a committed script, chain, or reproducibility manifest; nothing is quoted from memory. The exact-amplitude derivation (Sec. II) and the bounce-transmission bound (Sec. III) are committed, pinned to commit `68309c8`, at

https://github.com/Hubify-Projects/bigbounce/tree/68309c8/research/theory_audit

https://github.com/Hubify-Projects/bigbounce/tree/68309c8/research/cubic_bounce_transmission

respectively. The bounce-to-local shape overlap ($r$, Sec. VI) is imported from this lab's companion Fisher-forecast draft at `research/focused_paper_source_integration/` and `outputs/survey_reach_fnl.json`. The three channel computations of Secs. IV–VI are committed at `research/track_a3_multichannel/`: `pta_gamma_reproduce.py`, `pbh_abundance_fnl.py` (first-pass Press–Schechter, retained for comparison only), `pbh_compaction_fnl.py` (compaction-function criterion, this paper's principal PBH result), `survey_reach_fnl.py`, and `pta_injection_30bin_2026_09_02.py` (the Sec. IV C injection-recovery validation), with results in `outputs/*.json` and manifests at `reproducibility/manifests/experiments/a3-*.json` (PTA/PBH/reach, including `a3-pbh-compaction-fnl.json`), `reproducibility/manifests/experiments/p2-fnl-*.json` (exact-amplitude derivation and validation), and `reproducibility/manifests/experiments/p2-a2-*.json` (bounce-transmission computation). The pulsar-timing chain is `pipelines/p3_pta_mcmc/free_spectrum_real_2026-05-01/chain_real_freespec.npy` (SHA-256, in two halves to allow line-breaking: `50abc38a04e1bf886adc833b5eb653be4e986877fe2` `476f87a78927f4f3610fc`), derived from the NANOGrav 15-yr KDE free spectra, Zenodo `10.5281/zenodo.8060824`. This work's own code and artifacts are pinned by the GitHub commit hash above rather than a frozen-release DOI of their own; minting one is a maintainer action planned for the packaging stage. The exact-amplitude theory lineage this paper builds on is archived at this lab's companion P2 Zenodo record, `10.5281/zenodo.21461881`. Every computation reported in this paper ran on a local CPU (Apple M5, 24 GB, macOS 26.5) at zero compute cost; the PTA/PBH/reach/injection channels ran in under 6 s total. Compute venue, wall-clock time, and external data sources are recorded per-experiment in the manifests cited above, consistent with this lab's standing reproducibility policy (directive Q2).

## AI USAGE DISCLOSURE

Symbolic derivations reported in Secs. II–III were performed by deterministic sympy scripts written and executed under the author's direction; every intermediate result is printed by the committed scripts and independently checked against the cited literature limits before use. The channel computations of Secs. IV–VI were performed by committed Python scripts against real data and cited published values, not by an LLM. Claude (Anthropic) assisted with literature cross-checks, prose drafting, and LaTeX formatting under the author's direction; all physics content, equations, and numerical results were verified by the author against the committed script output.


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